

Eucalypt growth in monoculture and silvopastoral systems with varied tree initial densities and spatial arrangements

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Abstract Brazil has large stretches of degraded land, mainly occupied by pasture. Recently, the Brazilian government created a program providing financial support to recover these lands throughout the country by establishing agroforestry systems, among others. To select the most appropriate initial spatial arrangements and tree planting density for such systems, the plant growth and leaf area index (LAI) of a *Eucalyptus camaldulensis* clone were evaluated in five spatial arrangements: 3.6×2.5 m, and 3.3×3.3 m (monoculture), and $(2 \times 2) + 10$ m, $(3 \times 3) + 9$ m, and 9×3 m (silvopastoral systems), with 1,111, 918, 833, 556 and 370 plants per hectare, respectively. The arrangement did not affect plant height growth ($p > 0.05$) up to 50 months. The average diameter was affected by plant proximity in the planting line, being smaller ($p < 0.05$) in the arrangements $(2 \times 2) + 10$ m, and 3.6×2.5 m, and by the number of lines in the rows, as the plants in the single line arrangement (9×3 m) attained the highest diameter. Individual volumes of high initial density

stand were lower than in the silvopastoral systems. The arrangements $(2 \times 2) + 10$ m, 3.6×2.5 m, and 3.3×3.3 m did not differ ($p > 0.05$) in volume per hectare, which was higher ($p < 0.05$) than for the other arrangements studied. LAI decreased with age and it was higher ($p < 0.01$) in the monoculture as compared to the silvopastoral systems. The silvopastoral system can be established in the $(2 \times 2) + 10$ m arrangement for high biomass production besides having the additional benefit of financial returns from the pasture, as compared with the monoculture. The 9×3 m arrangement is preferable for large diameter log production.

Keywords Initial tree spacing · Tree spatial arrangement · Eucalypt clone plantation · Leaf area index · Silvopastoral system

Introduction

Extensive eucalypt plantations have been established in Brazil, mainly in the savannah biome. Initially, these plantations showed low productivity due to inadequate management (Stape et al. 2010). By adopting site-specific genotype, minimum cultivation, fertilization and other techniques, the average productivity of the eucalypt forests established in the monoculture ($6\text{--}12\text{ m}^2$ per plant) increased to about

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40 m³ ha⁻¹ year⁻¹ (Gonçalves et al. 2004; ABRAF 2013), although an even higher productivity can be achieved with adequate fertilization and irrigation (Stape et al. 2010). The eucalypt has been established in monoculture in more than 4.5 million hectares (ABRAF 2013) and, has also been adopted in the agroforestry systems in Brazil (Dubè et al. 2002; Oliveira Neto et al. 2010; Fontan et al. 2011; Cubbage et al. 2012) due to its high productivity and wide variety of genotypes and multiproducts.

The savannah biome, extending across 205 million hectares in central Brazil, is characterized by highly weathered, acidic soils, having low natural fertility, and which has limited cash crop production (Thomas et al. 2000). Cattle rearing in the native and planted pastures were the chief activity in this region although because of low technology, such as overgrazing and the lack of fertilization, intense land degradation resulted (Costa and Rehman 2005). Today, the Brazilian savannah region plays an important role in grain production, cattle rearing and reforestation due to the increasing use of advanced production techniques (Thomas et al. 2000; Gonçalves et al. 2004); however, much is still to be done to recover all the degraded land.

Currently, there is a high demand for more sustainable production models, such as the agroforestry systems (Cubbage et al. 2012). These systems are important to enhance soil fertility (Jackson and Ash 1998; Pinho et al. 2012), increase the farm economical return (Dubè et al. 2002; Souza et al. 2007; Prasad et al. 2010; Pacheco et al. 2013), and sequester the carbon that can be stored in the tree biomass and in the soil (Montagnini and Nair 2004; Haile et al. 2010; Howlett et al. 2011; Tonucci et al. 2011; Jose and Bardhan 2012), among others. Therefore, the agroforestry systems and techniques such as no-tillage were included in the Brazilian government financing program called low carbon agriculture to enable the recovery of degraded pasture and agricultural land (Stabile et al. 2012).

The silvopastoral system has been considered a good option for cattle-rearers to recover degraded pasture (Dubè et al. 2002) as the trees in the pasture may improve the forage nutritional quality (Jackson and Ash 1998; Carvalho et al. 2002; Tripathi et al. 2005; Souza et al. 2010) and animal comfort (Baliscei et al. 2013). Tree cover can increase forage biomass (Souza et al. 2010) or decrease it, if shading is too high

(Jackson and Ash 1998; Rozados-Lorenzo et al. 2007; Paciullo et al. 2011). Thus, LAI is a helpful tree canopy characteristic that can be used to predict pasture potential production. Rozados-Lorenzo et al. (2007) reported that the arboreal component leaf area index was inversely correlated with annual pasture production. LAI, therefore, should be maintained at an optimal level to reduce the detrimental effect of the trees on forage production, due to shading. A suitable LAI can be obtained by selecting genotypes with low crown density, adequate tree spatial arrangements and initial planting densities, supported by pruning and thinning, among others, to allow sufficient radiation transmittance (Ares and Brauer 2005; Chaves et al. 2007; Oliveira Neto et al. 2010; Fontan et al. 2011).

The definition of tree planting density, spacing and arrangement should consider genotype, site carrying capacity, forest use and the growth resource requirements of the trees, crops and or, pasture to be established in the consortium (Binkley et al. 2004; Gonçalves et al. 2004; Kruschewsky et al. 2007; Magalhães et al. 2007; Rozados-Lorenzo et al. 2007; Oliveira Neto et al. 2010; Prasad et al. 2010; Salles 2010; Silveira et al. 2011). By increasing the tree planting density, greater competition is created among the plants, resulting in an increase in the number of trees with reduced diameter (Clutter et al. 1983; Leite et al. 2006; Kruschewsky et al. 2007). Low tree planting density encourages a healthy association with the crops and, or pasture (Dubè et al. 2002; Prasad et al. 2010; Oliveira Neto et al. 2010) and promotes growth in diameter and individual volume, although with low yield (Pinkard and Neilsen 2003; Leite et al. 2006). This increase in tree size may give greater financial returns if the trees are managed through pruning and thinning to obtain higher wood quality (Ares and Brauer 2005; Fontan et al. 2011; Forrester and Baker 2012; Nogueira et al. 2005). Therefore, the production of the tree component and, or the crop and pasture should be considered as a whole, in order to define the most adequate spatial arrangements and initial tree planting density of the arboreal component in an agroforestry system.

The present study focused on the growth, yield and LAI of a *Eucalyptus camaldulensis* natural hybrid clone established in the monoculture and silvopastoral systems, under different initial tree planting density and spatial arrangements, focusing on the following questions: (a) when growth resources are limited, is

tree growth favored if established in the silvopastoral system, with one or two planting lines per tree row, having shorter distances between the plants in the rows and large open land spaces between the rows?; (b) is the decrease in LAI noted in the silvopastoral arrangements when compared with the monoculture ones?; (c) are the silvopastoral systems, established with eucalypt, an effective method for cattle-rearers, to obtain additional financial returns from the forest?

Materials and methods

Study area

The study was conducted at Votorantim Siderurgia S.A, Vazante, MG (17°36′09″S and 46°42′02″W–550 m altitude). The original vegetation of the study area is cerrado (savannah) which was replaced by *Brachiaria brizantha* (Hochst. ex A. Rich.) Stapf cv. Marandu pasture. The regional average annual rainfall is 1,312 mm (1989–2009), with an average monthly rainfall of 222 mm, from November to March, and 29 mm, from April to October. The average annual, mean maximum and mean minimum temperatures are 26.0, 35.0 and 18.0 °C, respectively, whereas the average annual relative humidity is 74 %. The potential evapotranspiration of 1,689 mm and water deficit of 497 mm (mostly from April to October) were estimated by the Thornthwaite method (D’Angiolella and Vasconcellos 2004). The soil type in the area is classified as oxisol with very low natural fertility.

Soil samples of the whole experimental area were taken at depths of 0–20 and 20–40 cm to determine the physical and chemical characteristics prior to starting the experiment (Table 1). These results show 51.5 % sand, the soil being classified as sandy-textured. As the soil pH and the cation exchange capacity are very low, and the aluminum saturation is high, this is considered a low fertile soil.

The experiment

A pasture land established with *B. brizantha* (Hochst. ex A. Rich.) Stapf cv. Marandu, overgrazed for decades, was planted with a *E. camaldulensis* natural hybrid in the monoculture and silvopastoral systems. No crop

plantation was introduced in this system in order to simulate the conditions normally adopted by the cattle-rearers. The treatments included the following eucalypt tree arrangements: 3.6 × 2.5 m (T1), 3.3 × 3.3 m (T2) (monoculture) and, (2 × 2) + 10 m (T3), (3 × 3) + 9 m (T4) and 9 × 3 m (T5) (silvopastoral system). T3 and T4 were established with double lines of eucalypt in the 2 × 2 m or 3 × 3 m arrangement, with 10 or 9 m width being the distance between the rows, respectively. The corresponding densities (trees ha⁻¹) are: 1,111 (T1), 918 (T2), 833 (T3), 556 (T4) and 370 (T5), respectively. The experimental design was completely randomized due to the high homogeneity of the relief (slope of 1–2 %) and soil type, with three replicates for each tree spatial arrangement, 15 plots in total (experimental units). The plots were established side by side. Each plot had 24 plants (six lines of four plants each) with one border line; however, their size varied with the tree arrangements as a fixed number of trees was considered per plot.

Subsoiling was done to a depth of 30–40 cm, and 743 kg ha⁻¹ of natural reactive rock phosphate was applied along this line before planting. The trees were planted in December 2006, followed by the application of 130 kg ha⁻¹ of NPK (10:28:06) with 0.6 % of B and 0.4 % of Zn. Two months after planting, 95 kg ha⁻¹ of ammonium sulfate was applied after chemical weeding, with glyphosate. Four months after planting, 106 kg ha⁻¹ of ammonium sulfate and 25 kg ha⁻¹ of borogran (10 % of B) were applied. The fertilization per plant varied according to the number of trees in each spatial arrangement, as a fixed amount was applied on a hectare basis. As the logistic to introduce cattle in a small area is quite complex, mechanical weeding was adopted whenever required. Chemical weeding was adopted only up to 1 m apart from the tree planting line, to enable maintenance of the pasture between the rows until the trees were harvested.

Data collection and analysis

In order to assess tree growth over an age sequence, the diameter at 1.3 m height (dbh) and total height (Ht) of all the trees in each plot were measured at 10, 18, 23, 33, 38 and 50 months, respectively, after tree planting. Dominant height was based on the five tallest trees in each plot and age.

Table 1 Physical and chemical soil properties of the area studied, for the depths of 0–20 cm and 20–40 cm. Soil samples were obtained in the *Brachiaria brizantha* pasture before the

establishment of the experiment in the savannah region, in Vazante, Minas Gerais State, Brazil

Soil depth (cm)	Clay (%)	Silt (%)	Sand (%)	pH (water)	H+Al (cmol _c kg ⁻¹)	Al (cmol _c kg ⁻¹)	Ca (cmol _c kg ⁻¹)	Mg (cmol _c kg ⁻¹)	K (cmol _c kg ⁻¹)	CEC
0–20	26	24	50	5.1	4.0	0.3	1.1	1.0	0.45	1.0
20–40	24	23	53	4.8	4.0	1.2	0.3	0.3	0.21	0.7
Soil depth (cm)	P (mg kg ⁻¹)	S (mg kg ⁻¹)	OM (%)	V (%)	M (%)	B (mg kg ⁻¹)	Zn (mg kg ⁻¹)	Fe (mg kg ⁻¹)	Mn (mg kg ⁻¹)	Cu (mg kg ⁻¹)
0–20	1.0	2.0	3.0	39	10	0.1	4.7	109	13.9	1.2
20–40	0.7	1.3	1.5	17	60	0.1	0.8	78	3.0	1.1

P, K, B, Zn, Fe, Mn and Cu Mehlich I, S monobasic calcium phosphate, B hot water, OM calorimetric method, V (sum of base = K + Ca + Mg + Na), M aluminum saturation, CEC cation-exchange capacity

At 50 months, three trees each per 2 cm diameter class were sampled for scaling in each replication of the following silvopastoral arrangements: (2 × 2) + 10 m, (3 × 3) + 9 m, and 9 × 3 m. The diameter with bark was measured at ground level, and at 0.5, 1.0, 2.0, 4.0 m, and so on, to every 2 m, to the height where the trunk diameter was 4.0 cm. The volume with bark was determined for these sampled trees using Smalian's Formula. To estimate the tree individual volume (V_i), per age, the model of Schumacher and Hall (1933) was fitted, in the logarithmic form, by the least square ordinary method:

$$\ln V_i = \beta_0 + \beta_1 \ln dbh + \beta_2 \ln Ht + \ln \varepsilon_i \quad (1)$$

where V_i is the volume per tree; β_0 and β_1 is the model parameters; ε is the model error.

Equations to estimate the individual tree volume, per age, for the 3.6 × 2.5 m (T1) and 3.3 × 3.3 m (T2) arrangements (monoculture) were provided by Votorantim Siderurgia based on the commercial plantation inventory of the same genotype, soil and cultural treatments.

The Gompertz (2) and Logistic (3) models were both fitted for each spatial arrangement for the following variables: diameter, total and dominant height, basal area, and volume per tree and per hectare. These models were fitted by the Levenberg–Marquardt method using Statistica software (StatSoft, Inc. 2012).

To define whether the Gompertz or Logistic model was the best for each variable, an analysis of the correlation coefficient was performed between the observed and estimated values, $R_{\hat{y}y}$, and the residual

standard error (S_{yx}) (Ratkowsky 1990; Campos and Leite 2013). The equations obtained for each tree initial arrangement for each variable were compared using the identity tests for nonlinear models (Bates and Watts 1988; Regazzi 2003; Regazzi and Silva 2004, 2010).

$$Y = \alpha e^{\beta - \gamma A} + \varepsilon \quad (2)$$

$$Y = \frac{\alpha}{1 + \beta e^{-\gamma A}} + \varepsilon \quad (3)$$

where Y is the dependent variable; α , β , γ is the model parameters; A is the age, in months; ε is the model error; α is the Y value when A tends to infinity, i.e., α is the model asymptotic value (Drapper and Smith 1981).

The current annual increment (CAI) and the mean annual increment (MAI) were obtained for volume per hectare for each spatial arrangement. CAI is the increment in Y values (m³ ha⁻¹) which a tree puts on in a year period. MAI is the average production (m³ ha⁻¹ year⁻¹) from planting time to a specified age.

The two-parameters Weibull function (Murthy et al. 2004) were fitted to predict the diameter distribution at the age of 50 months, for each tree's initial arrangement. The adjusted distributions were compared using the statistical procedure of Leite and Oliveira (2002) with a 5 % level of significance. The statistical procedure proposed, resulted from a combination of the statistic F as modified from Graybill (1976), the test of the medium error and, the linear correlation coefficient. Based on these data, a ruling decision was proposed to test the hypothesis of

identity between two analytical methods or any two vectors, that is, groups of quantitative data. Termed Y_1 and Y_j , the two vectors of quantitative data are represented as follows:

$$y_1 = \begin{bmatrix} 1 & Y_{11} \\ 1 & Y_{12} \\ \vdots & \vdots \\ 1 & Y_{1n} \end{bmatrix}; \quad y_j = \begin{bmatrix} Y_{j1} \\ j_{j2} \\ \vdots \\ j_{jn} \end{bmatrix}$$

the relationship between y_1 and y_j can be matrically expressed for $y_j = y_1\beta + \varepsilon$, where:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}; \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_3 \end{bmatrix}$$

Under the normality hypothesis

$$F(H_0) = \frac{(\beta - \theta)'(y_1'y_1)(\beta - \theta)}{2QMR} \sim F_{\alpha(2, n-2 \text{ g.l.})},$$

$$\beta = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}, \quad \theta = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad y_1'y_1 = \begin{bmatrix} n & \sum Y_1 \\ \sum Y_1 & \sum Y_1^2 \end{bmatrix}$$

in which RMS is the residual mean square of the regression $Y_j = \beta_0 + \beta_1 Y_1$. Thus, with $n-2$ degrees of freedom and at significance level α these statistics can be used to test the hypothesis $H_0: \beta = [0 \ 1]$. If $F(H_0) \geq F_\alpha(2, n-2 \text{ d.f.})$ the hypothesis is rejected. On the other hand, H_0 and $Y_j = Y_1$ are not rejected, accepting the identity between the two methods appraised. Considering that the errors $e_i = (Y_{ji} - Y_{1i})/Y_{1i}$ follow a normal distribution and that j indicates the alternative method and 1 the standard method, the hypothesis $H_0: \bar{e} = 0$ can be tested, using the statistic t , given by $t_{\bar{e}} = \frac{\bar{e}-0}{S_{\bar{e}}}$; in which $S_{\bar{e}} = \frac{S_e}{\sqrt{n}}$ with $n-2$ degree of freedom. If $t_{\bar{e}} \geq t_\alpha(n-1 \text{ d.f.})$, this hypothesis is rejected. In summary, Y_j and Y_1 (two tree spatial arrangements) are identical when $p > 0.05$ for F and t tests and, $r_{Y_j Y_1} \geq 1 - |\bar{e}|$.

At 38 and 50 months after planting, one set of measurements of the LAI was obtained with two LAI-2000 canopy analyzer sensors for each plot: one was set in an open area near the experimental area while the other one was used to take the readings in the plots, at dawn or late afternoon. LAI readings were obtained from a zenithal angle of 68 degrees. In the 9×3 m arrangement, readings were taken at three points: the first one was located between two trees in the planting

line and, the second and third ones, were at $\frac{1}{4}$ and $\frac{1}{2}$ of the greatest distance between the tree rows. In the arrangements $(2 \times 2) + 10$ m and $(3 \times 3) + 9$ m, the first measurement was taken at the diagonal intersection of four plants in the double tree rows, followed by three other points, as in the 9×3 m arrangement. The LAI readings were taken at the intersection of the diagonals of four plants in the plots for the 3.6×2.5 m and 3.3×3.3 m arrangements.

A one-way analysis of variance was performed to test the LAI in different spatial arrangements, for each age. Least significant difference by Tukey's test was performed at 1 % level of significance. Analyses were performed using the software Statistica 10.0 (Statsoft Inc. 2012).

Results

After testing both the Gompertz and logistic models for all the variables studied, it was observed that the Gompertz model was the better to describe the growth in diameter, total height and basal area per hectare; also, the logistic model better described the dominant height and volume per plant and per hectare.

The identity among the tree spatial arrangement growth models was tested for each variable and the final equations presented in Table 2. When two or more spatial arrangement growth equations of a given variable were similar ($p > 0.05$), a common equation was obtained and their correlation coefficients were mostly greater than 0.94 ($p < 0.05$).

The tree spatial arrangements did not influence ($p > 0.05$) the total height or the dominant height growth; therefore, only one equation was used to describe these variables (Table 2; Fig. 1a, b).

On the other hand, there was a significant effect of the initial tree spatial arrangement on all the other variables studied (Table 2; Fig. 1).

For diameter growth, the 9×3 m arrangement presented the highest ($p < 0.05$) asymptotic value (α parameter) (Table 2), achieving maximum growth at older ages when compared with the other arrangements. The growth curves of the arrangements 3.6×2.5 m and $(2 \times 2) + 10$ m were similar, with a dbh asymptotic value 31.6 % much less than that observed for the 9×3 m arrangement. The diameter curves of the 3.3×3.3 m arrangement did not differ

Table 2 Estimated parameters for the Gompertz (total height, diameter at 1.3 m and basal area) or Logistic (dominant height, volume per tree and volume per hectare) models, as a function of age (months), and corresponding correlation coefficients between the estimated and observed values ($R_{\hat{y}y}$) and residual

standard error (S_{yx}) of a *Eucalyptus camaldulensis* clone, under different spatial arrangements, in the savannah region, Vazante, Minas Gerais State, Brazil, where T1 = 3.6×2.5 m; T2 = 3.3×3.3 m; T3 = $(2 \times 2) + 10$ m; T4 = $(3 \times 3) + 9$ m; T5 = 9×3 m

Treatments	Gompertz model parameters			$R_{\hat{y}y}$	S_{yx}	Treatments	Logistic model parameters			$R_{\hat{y}y}$	S_{yx}
	α	β	γ				α	β	γ		
Height (m)						Dominant height (m)					
All	27.9229	1.2157	0.0510	0.97	1.44	All	29.1794	12.1794	0.0833	0.97	1.49
Diameter (cm)						Volume (m ³ tree ^{−1})					
T1;T3	16.2084	1.2406	0.0797	0.96	1.10	T1	0.1244	53.6035	0.1247	0.91	0.01
T2;T4	18.3513	1.3601	0.0826	0.98	0.78	T2	0.1863	154.6736	0.1458	0.97	0.14
T5	21.3343	1.3960	0.0759	0.99	0.44	T3	0.2020	459.6863	0.1726	0.97	0.14
						T4	0.2353	128.2483	0.1541	0.99	0.00
						T5	0.3169	248.3599	0.1460	0.99	0.00
Basal area (m ² ha ^{−1})						Volume (m ³ ha ^{−1})					
T1;T2;T3	22.9968	1.6469	0.0607	0.94	2.08	T1;T2;T3	145.3105	99.2796	0.1316	0.97	10.87
T4	12.8777	2.0449	0.0790	0.96	2.08	T4;T5	109.8492	142.3333	0.1415	0.95	10.80
T5	14.0891	1.9141	0.0638	0.98	0.57						

Growth equations representing two or more spatial arrangements indicate that there are no differences ($p > 0.05$) among them, by applying the identity test

from those of the $(3 \times 3) + 9$ m one, both having about the same distance between the planting lines.

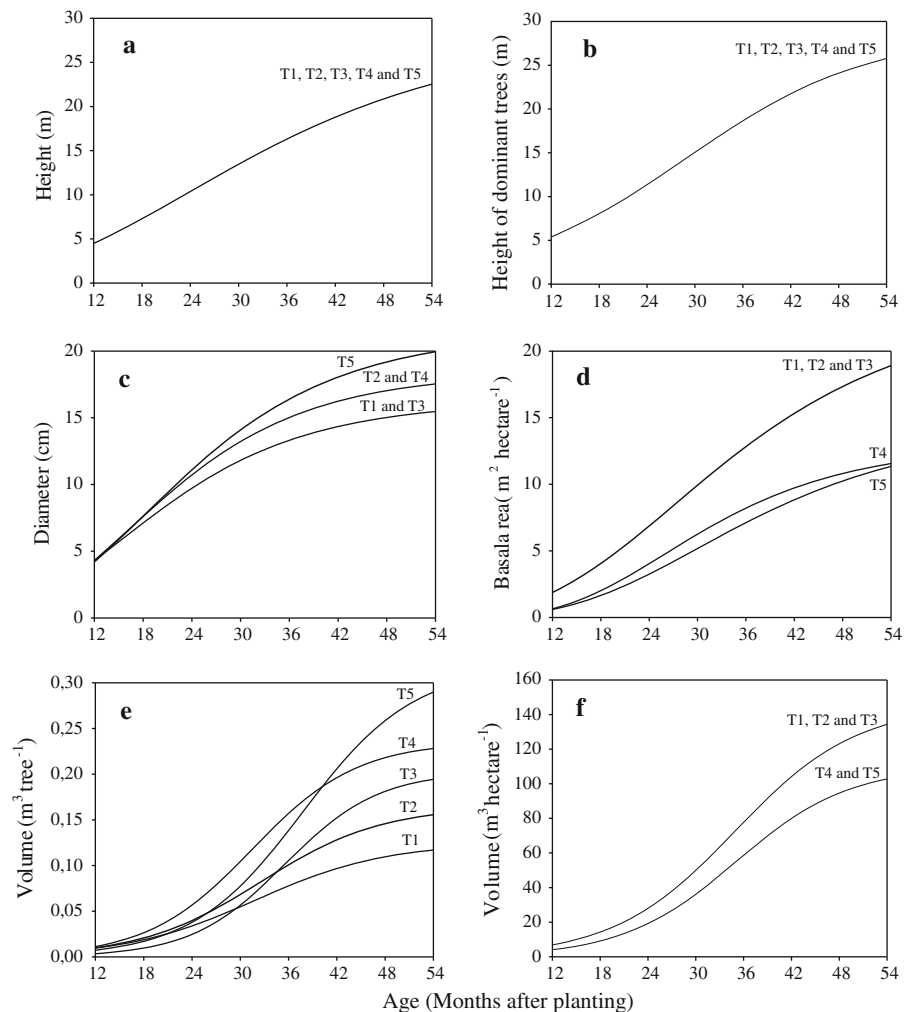
Based on the Leite and Oliveira (2002) statistical test, the adjusted Weibull diameter distribution at the age of 50 months differed among the spatial arrangements (Fig. 2). An evident increasing negative skewness was observed with decreasing tree initial density. The arrangement 9×3 m, with only one planting line, presented the lowest number of suppressed trees and the greatest number of trees in the highest diameter classes. The arrangement $(3 \times 3) + 9$ m also presented a low number of non-dominant trees. Therefore, it can be observed that the distribution of trees with the largest diameter is more similar to the $(2 \times 2) + 9$ m than to the 9×3 m arrangement. The statistical test used in the analysis of the diameter distribution indicates that the particular diameter at which there were more than 37 % of the trees (beta-coefficient) for each arrangement were 15.7 (3.6×2.5 m), 16.6 (3.3×3.3 m), 17.7 [$(2 \times 2) + 10$ m], 18.6 [$(3 \times 3) + 9$ m] and 20.8 (9×3 m), respectively. The treatments with the shortest distance between the plants in the planting line (2 or 2.5 m)

presented a higher number of trees in the smallest diameter classes.

The tree individual volume (Table 2; Fig. 1e) differed ($p < 0.05$) among all the tree initial arrangements and increased with the decrease in stand density. The asymptotic values of the 9×3 m arrangement were 155 % greater than that for the 3.6×2.5 m, and 82 % greater than for the $(3 \times 3) + 9$ m arrangement (Table 2). The $(2 \times 2) + 10$ m presented characteristics much closer to the monoculture arrangement than to the 9×3 m arrangement.

No differences ($p > 0.05$) were observed in the basal area (Table 2; Fig. 1d) and volume per hectare (Table 2; Fig. 1f) among the 3.6×2.5 m, 3.3×3.3 m and $(2 \times 2) + 10$ m arrangements. The asymptotic values of the basal area and volume per hectare for these treatments were, 63.2 and 38.3 % respectively, greater than for the 9×3 m treatment (Table 2). The MAI and the CAI of these three arrangements were greater ($p < 0.05$) than for those with the lower density and the technical harvesting age was about 48 months for these tree arrangements (Fig. 3). The 9×3 m and the $(3 \times 3) + 9$ m differed ($p < 0.05$)

Fig. 1 Total height (a), dominant height (b), diameter at 1.3 m (c), basal area (d), volume per tree (e) and volume per hectare (f) curves, as a function of age (months), for a *Eucalyptus camaldulensis* clone, under different spatial arrangements, in the savannah region, Vazante, Minas Gerais State, Brazil, where T1 = 3.6×2.5 m; T2 = 3.3×3.3 m; T3 = $(2 \times 2) + 10$ m; T4 = $(3 \times 3) + 9$ m; T5 = 9×3 m. The growth curves representing two or more spatial arrangements indicate that there were no differences ($p > 0.05$) among them, by using the models identity test. The equations used to draw these curves are seen in Table 2



in the basal area, although they did not differ for volume per hectare, and presented the lowest values.

The higher LAI values at 38 and 50 months were attained in the stands with 1,111 (T1) and 918 (T2) trees ha^{-1} , with a sharp decline in the LAI for these two treatments between 38 and 50 months. The arrangements in the silvopastoral systems (T3, T4 and T5) presented a low LAI and a smaller difference between 38 and 50 months (Fig. 4).

Discussion

The lack of effect of the tree planting spatial arrangements on the average total height of the stand or the dominant height was also observed in other studies (Leite et al. 1997; Khan and Chaudhry 2007;

Magalhães et al. 2007). Leles et al. (2001) reported average height reduction only in the very high stand density (3.0×1.5 m and 3×1 m) of the 52-month-old *E. camaldulensis* and *Eucalyptus pellita* stands. Gerrand and Neilsen (2000) reported a decrease in the total height of *Eucalyptus nitens* in very low or very high stand densities. The reduced effect of the initial spatial arrangement on dominant height makes this variable very useful to indicate site quality (Clutter et al. 1983; Campos and Leite 2013). In the present study, the fertilizer applied per plant increased on a per plant basis with the decrease in tree planting density. Even with this difference in fertilization per plant, the dominant height did not vary.

Diameter growth is affected by the available space per plant, i.e., by the availability of growth resources (Leles et al. 2001; Oliveira Neto et al. 2003; Nogueira

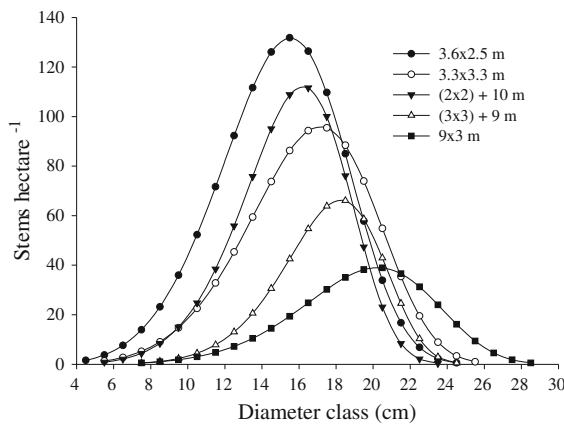


Fig. 2 The number of stems per hectare per class of 1 cm diameter, estimated by the two parameters Weibull function, of the 50-month-old *Eucalyptus camaldulensis* clone, under different spatial arrangements, in the savannah region, Vazante, Minas Gerais State, Brazil. The adjusted distributions were compared by statistical procedure of Leite and Oliveira (2002) and all the arrangements were found to be different ($p < 0.05$)

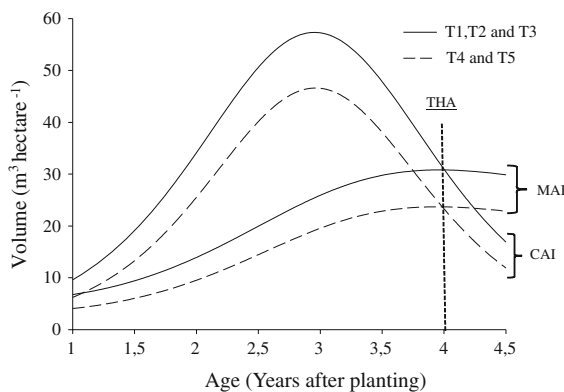


Fig. 3 Current annual increment (CAI), mean annual increment (MAI) and technical harvesting age (THA) of a *Eucalyptus camaldulensis* clone, as a function of age, under different spatial arrangements, in the savannah region, Vazante, Minas Gerais State, Brazil. Where: T1 = 3.6×2.5 m; T2 = 3.3×3.3 m; T3 = $(2 \times 2) + 10$ m; T4 = $(3 \times 3) + 9$ m; T5 = 9×3 m. THA is obtained when the CAI=MAI. The CAI and MAI curves when representing two or more spatial arrangements indicate that there were no differences ($p > 0.05$) among them by using the models identity test

et al. 2005; Khan and Chaudhry 2007; Magalhães et al. 2007). The variation in space per plant can be obtained by varying the initial tree planting density, tree spatial arrangement and thinning. Proximity among the plants probably explains the greater diameter growth in the single row (9×3 m) when compared with the $(3 \times 3) + 9$ m treatment, as in the double row

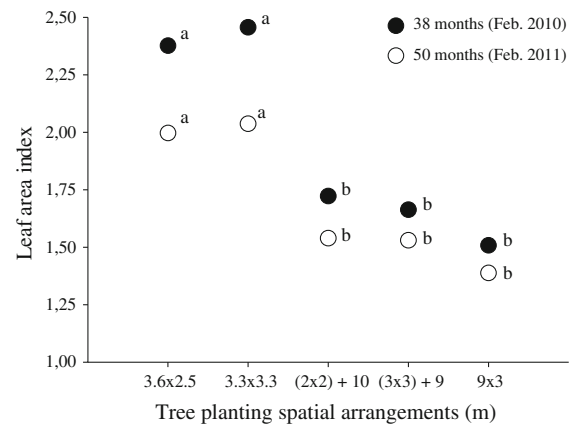


Fig. 4 The LAI of a *Eucalyptus camaldulensis* clone stand at the ages of 38 (closed circle) and 50 (open circle) months, as a function of spatial arrangements, in the savannah region, in Vazante, Minas Gerais State, Brazil. Significant differences (Tukey test; $p < 0.01$) among the arrangements are indicated by different letters

arrangement plant competition is prevalent in both the planting line as well as between the planting lines. The distance between the plants in the planting line appears to influence diameter growth, as the tree diameter in the $(3 \times 3) + 9$ m arrangement (silvo-pastoral system) was similar to the 3.3×3.3 m (monoculture) and, both were greater than for the $(2 \times 2) + 10$ m, and 3.6×2.5 m, arrangements, with lesser distance between the plants in the planting line. An increase in the diameter with the increasing distance between the plants in the planting line was also observed for the *E. camaldulensis* $\times$ *Eucalyptus urophylla* clone stand (Kruschewsky et al. 2007).

Despite substantial differences in the tree planting densities and arrangements among the 3.6×2.5 m, 3.3×3.3 m and $(2 \times 2) + 10$ m treatments, the identity test showed no distinction in their basal area curves. The average diameters of the 3.6×2.5 m and the $(2 \times 2) + 10$ m arrangements showed the lowest values, but the proportion of trees in the higher diameter classes was greater than for the 3.3×3.3 m (Fig. 3), increasing their basal area. The tree competition in the inter-row of the $(2 \times 2) + 10$ m arrangement is reduced, and, at older ages, the trees may attain a greater diameter increment. Consequently, the basal area in this arrangement may be different from the others, as it varies with the existence of trees of larger diameter (Campos and Leite 2013). The basal area of the 9×3 m arrangement, with a low initial tree

planting density ($370 \text{ trees ha}^{-1}$), was only slightly lower than that of the $(3 \times 3) + 9 \text{ m}$ ($556 \text{ trees ha}^{-1}$) because the diameter was higher in the $9 \times 3 \text{ m}$ stand, favoring the increase in the basal area. The greater basal area in the $9 \times 3 \text{ m}$ arrangement stand (single line) as compared with the $(3 \times 3) + 10 \text{ m}$ one (double line) was also observed for the 51-month-old *E. camaldulensis* $\times$ *E. urophylla* clone stand (Oliveira et al. 2009).

The individual tree volume and its increment were higher in the single or double line arrangements, as the distance between the plants favored their individual volume or biomass as demonstrated in other studies as well (Bernardo et al. 1998; Leles et al. 2001; Kruschewsky et al. 2007; Oliveira et al. 2009). Minor differences in diameter may explain the difference in individual tree volume between the arrangements due to the diameter squared effect. High tree densities generate non-dominant trees with low individual tree volume (Leite et al. 2006; Oliveira et al. 2009).

The rectangularity, a ratio between the greatest and the least distances between the plants, may interfere in the availability of growth resources (Pinkard and Neilsen 2003; Gonçalves et al. 2004). Higher rectangularity values for the initial planting arrangements indicate that the plants are much closer, for example, in the planting line, when compared with the distance between the rows. The arrangements with high rectangularity is normal in the agroforestry systems, even when more than two lines per row is adopted, maintaining a greater distance between the rows to favor the consortium with the crops and pasture (Ares and Brauer 2005). Gerrand and Neilsen (2000) reported no effect of rectangularity, varying from 1 to 3, with regard to the diameter and height growth of *E. nitens*. In the present study, the $3.3 \times 3.3 \text{ m}$ arrangement (rectangularity equal to 1) favored higher individual volume compared with the $3.6 \times 2.5 \text{ m}$ (rectangularity of 1.44). Both the arrangements presented lower individual tree volume when compared with the silvopastoral arrangements studied. However, rectangularity should not be analyzed alone, because individual tree growth is also influenced by the number of neighboring trees. It is likely that a greater distance between the rows, in the single or double row arrangement, with a lesser number of neighboring plants, may benefit plant growth, by facilitating access to the growth resources.

The final wood production is an important component in the analysis of the agroforestry systems or

monoculture. Tree growth depends upon resource availability, which can be managed by using different initial tree planting densities and or spatial tree arrangements. If agroforestry systems are adopted, one should also consider the potential production of the crop or pasture to be planted in the inter-row. Normally, volume or wood biomass per hectare decreases with a reduction in the tree planting density (Bernardo et al. 1998; Leles et al. 2001; Kruschewsky et al. 2007; Magalhães et al. 2007; Oliveira et al. 2009). Similar to the results in the basal area, in the present study, the yield of the three arrangements with the higher initial tree planting density were similar, and greater than those with the low density due to the number of trees per hectare and the effect of tree arrangement on individual tree growth. The $9 \times 3 \text{ m}$ arrangement presented wood production similar to the $(3 \times 3) + 9 \text{ m}$ one, with the advantage of higher diameter growth, enhancing its management for multiproduct. The wood production of the two monoculture tree arrangements ($3.6 \times 2.5 \text{ m}$ and $3.3 \times 3.3 \text{ m}$) were similar to that of the silvopastoral system with the arrangement $(2 \times 2) + 10 \text{ m}$, which indicates that this last option offers a higher profitability as it allows for intercropping or pasture establishment. Kruschewsky et al. (2007) observed that the volume per hectare in a $3 \times 2 \text{ m}$ arrangement ($1,667 \text{ trees ha}^{-1}$) was only 16.4 % higher than in the $10 \times 2 \text{ m}$ ($500 \text{ trees ha}^{-1}$) 67-month-old *E. camaldulensis* $\times$ *E. urophylla* clone.

The tree roots and crown dynamics should also be considered in the analysis of the effect of the spatial tree arrangement on the competition among the trees and, between the trees and crops or pasture, mainly related to nutrients, water and light (Bernardo 1995; Leles 1995; Prasad et al. 2010; Oliveira Neto et al. 2010). Leles (1995) observed that the fine roots of the 52-month-old *E. camaldulensis* and *E. pellita* trees were well distributed over the total area in the $9 \times 9 \text{ m}$ arrangement, in the savannah region. Then, it is expected that, in the present study, competition for nutrients and water between the trees and grasses may be high at the age of 50 months. Usually, the crops are established soon after the trees are planted (Dubè et al. 2002; Prasad et al. 2010), i.e., when root competition is still low.

To maintain crop/pasture high productivity, tree spatial arrangement and crown characteristics should favor photosynthetic active radiation (PAR) transmittance (Prasad et al. 2010; Souza et al. 2010). LAI is a

good parameter to indicate the PAR transmittance through the forest canopy (Binkley et al. 2004; Chaves et al. 2007). Rozados-Lorenzo et al. (2007) reported that LAI can indicate the pasture potential production in the silvopastoral system. The LAI varies with genotype, age, growth resource availability and silvicultural treatments, such as pruning and thinning (Xavier et al. 2002; Silva-Pando et al. 2002; Smerthurst et al. 2003; Chaves et al. 2007; Fontan 2007; Boyden et al. 2008). Xavier et al. (2002) reported a LAI of 1.7–4.3 for *Eucalyptus grandis* hybrid plantation, with ages ranging from 12 to 84 months, in dense stands. *E. camaldulensis* and its hybrid is characterized by low density and narrow crowns, with pendant branches, delaying crown overlap and allowing higher radiation transmittance when compared with the other genotypes, especially those related to *E. grandis*, *E. urophylla* and *E. pellita* whose crowns are dense (Bernardo 1995; Xavier et al. 2002; Fontan et al. 2011).

The maximum LAI of an *E. camaldulensis* clone established in a 9.5×4.0 m arrangement, up to the age of 18 months, was lower than 0.47 and increased to 1.09 at the age of 30 months (Fontan 2007). In the present study, the LAI in the silvopastoral systems (Fig. 4) was higher than the values observed by Fontan (2007), probably due to the greater proximity among the trees, especially in the double-row arrangements. Taking into account that crops or pasture is established between the rows, the LAI should be maintained as low as possible. In Brazil, some farmers have planted trees, mainly eucalypt, in the pasture land, using more than three planting lines per row. The proximity among the trees increases with the number of tree planting lines in the row, which may increase the solar radiation interception by the tree canopy. In this case, the tree planting lines can be established in the east–west direction and, also, pruning and thinning should be applied to reduce the LAI and increase light transmittance (Chaves et al. 2007; Fontan et al. 2011).

The LAI decrease with age, as observed for all the tree silvopastoral arrangements in the present study, after canopy closure, has also been reported for other eucalypt stands (Xavier et al. 2002; Cacao 2008). The greatest difference in the LAI between the age of 38 months and 50 months observed for the arrangements with the highest tree planting density (monoculture) is a result of the high competition among plants, which may have promoted some leaf abscission.

The crop or pasture production in the inter-row varies with the distance from the trees (Prasad et al. 2010; Paciullo et al. 2011). Prasad et al. (2010) reported a lower cowpea yield in the first meter apart from the trees when compared with the center of the inter-row, these differences being greater when the inter-rows were greater than 7 m. In the agroforestry system established in the central savannah region in Brazil, the crops and pasture are established 1 m apart from the tree planting line to reduce competition between the trees and crops or pasture (Dubè et al. 2002). Paciullo et al. (2011) observed higher pasture benefit from tree shading in the distances of 7–10 m from the rows in a silvopastoral system in an arrangement of four planting lines (3×3 m) per row, with a distance of 30 m between the rows. These results indicate that the distance between the rows in the agroforestry systems depends on whether crops or pasture will be planted in the inter-rows.

The agroforestry systems with eucalypt, crops and or livestock has been considered profitable (Dubè et al. 2002; Souza et al. 2007; Prasad et al. 2010; Cubbage et al. 2012; Pacheco et al. 2013). As pointed out by Prasad et al. (2010), intercropping allows the annual income in the first years after the forest is established, when the costs are high, especially if the inter-rows are wide enough to maintain the high crop yield. A very large area of eucalypt in consortium with crops and pasture was established in the central savannah region of Brazil. If the system is to be managed for multiproduct, including sawmill, the net present value increases substantially when compared with energy wood production (Dubè et al. 2002; Souza et al. 2007). As eucalypt plants respond adequately to pruning and thinning in terms of growth and wood quality (Polli et al. 2006; Fontan et al. 2011, Forrester and Baker 2012) it is recommended to adopt these management methods to obtain higher financial returns. In this case, the harvesting rotation age would be delayed and the cattle-rearers can use the established pasture for longer periods.

In the present study, the eucalypt trees were established in a pasture already supporting *B. brizantha* without crop plantation, to simulate the system normally adopted by cattle rearers, particularly where mechanization is difficult. If this system is to be established in a highly mechanized area, it is recommended to establish cash crops in the first growing season after planting the trees, using legumes such as

soybean that are important in enhancing soil fertility and increase the financial returns of the system, as this crop is, today, a highly valuable commodity. Pacheco et al. (2013) reported regarding a system that has been adopted in Brazil, in which the crop, mainly maize, and *Brachiaria* sp. are sown at the same time in the first growing season after planting the trees and, 70 days after harvesting the maize the cattle are introduced into the system, allowing for a positive cash flow in this system, after the third year.

Conclusions

In Brazil, large extensions of degraded pasture that can be improved through the adoption of silvopastoral systems are present. If these systems are adequately established and managed, it is highly possible to obtain profitability with increased timber and pasture production, in addition to contribute towards sequestering and storing carbon, among other benefits. Some of the concerns to achieve success in such systems are related to the initial tree planting spacing and tree spatial arrangements, which will define the level of interaction between the trees or between the trees and grasses. Different products can be obtained from the silvopastoral system. A cattle-rearer would be looking for pasture with high quality and production, in addition to animal comfort. On the other hand, the forester would give priority to wood production. If sawn wood, plywood and poles are the major products needed, greater distances between the rows and between the plants in the planting line are required to obtain larger tree diameter. Also, pruning should be adopted in this situation, at a very early age of the stand, to avoid the development of thick branches. The 9×3 m tree spatial arrangement (single line) achieved the greatest stem diameter and individual tree volume, with the yield being similar to the $(3 \times 3) + 9$ m (double line). The highest yield from all the arrangements studied was obtained for the three highest initial density treatments, which include the two monoculture arrangements and the $(2 \times 2) + 10$ m arrangement, although with low tree diameter. These arrangements can be adopted for energy, pulp and other uses which depend on greater wood production. Considering that the monoculture and the $(2 \times 2) + 10$ m achieved the same production, this last arrangement would be more profitable, as

crops can be planted in the inter-rows during the earlier stages of forest establishment, followed by grass plantation for further cattle raising. This would allow farmers and ranchers to maintain their traditional activities, in conjunction with forestry activities. More than two tree planting lines should be avoided in the silvopastoral system, as the tree crowns became a barrier for light transmittance to the inter-row, reducing the pasture production. Crown and root dynamics and light transmittance should be considered in this kind of study to better understand the co-existence of trees and pasture. Finally, to reduce the detrimental effect of the trees on forage production due to shading, the genotypes should present low crown density and be established in adequate tree spatial arrangements allowing for sufficient radiation transmittance. Pruning and thinning should be adopted to reduce the LAI if the forest is to be managed for high quality poles or sawn wood, especially when two or more planting lines per row is the arrangement adopted.

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